

Problem 3 Solution.

let $x = \left| \frac{a+b-2c}{a+b+c} \right|$, let $y = \left| \frac{b+c-2a}{a+b+c} \right|$, and let $z = \left| \frac{a+c-2b}{a+b+c} \right|$

Since $(1+\frac{a}{b})(1+\frac{b}{c})(1+\frac{c}{a}) \geq 2$,

therefore $\frac{1}{abc}(a+b)(b+c)(a+c) \geq 2$

$\Rightarrow (a+b)(b+c)(a+c) \geq 2abc$

$\Rightarrow ((a+b+c)-c)((a+b+c)-a)((a+b+c)-b) \geq 2abc$

$\Rightarrow (a+b+c)^3 + (-a-b-c)(a+b+c)^2 + (ab+bc+ac)(a+b+c) - abc \geq 2abc$

$\Rightarrow (a+b+c)^3 - (a+b+c)^3 + (ab+bc+ac)(a+b+c) = 3abc$

$\Rightarrow (a+b+c)(ab+bc+ac) = 3abc$

$\Rightarrow a+b+c = \frac{3abc}{ab+bc+ac}$ (equation 1)

Then $x = \left| \frac{a+b-2c}{a+b+c} \right| \geq x = \left| \frac{a+b+c}{a+b+c} - \frac{3c}{a+b+c} \right| = \left| 1 - \frac{3c}{a+b+c} \right|$

using equation 1, $x = \left| 1 - \frac{\frac{3c}{ab+bc+ac}}{1} \right| = \left| 1 - \frac{ab+bc+ac}{ab} \right| = \left| -\frac{c(a+b)}{ab} \right|$

Similarly $y = \left| \frac{-a(b+c)}{bc} \right|$ and $z = \left| \frac{-b(a+c)}{ac} \right|$

assuming for the sake of contradiction that $x, y, z > \sqrt[3]{2}$

then $x \cdot y \cdot z > 2$

LHS = $\left| \frac{-a(b+c)}{bc} \right| \cdot \left| \frac{-b(a+c)}{ac} \right| \cdot \left| \frac{-c(a+b)}{ab} \right| = \left| \frac{-abc(a+b)(b+c)(a+c)}{(abc)^2} \right|$

(subbing in the values for x, y , and z)

$= \left| \frac{(a+b)(b+c)(a+c)}{abc} \right| \geq |2| \geq 2$ (since $(1+\frac{a}{b})(1+\frac{b}{c})(1+\frac{c}{a}) \geq 2$)

which contradicts the initial assumption of x, y , and z all being greater than $\sqrt[3]{2}$.

This means at least one of x, y, z must be less or equal to $\sqrt[3]{2}$.

Thus one of $\left| \frac{a+b-2c}{a+b+c} \right|, \left| \frac{b+c-2a}{a+b+c} \right|, \left| \frac{a+c-2b}{a+b+c} \right|$ must be less or equal

to $\sqrt[3]{2}$.